

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

Distributed Applications

University of Minho

2026-2027



Motivation and Goals

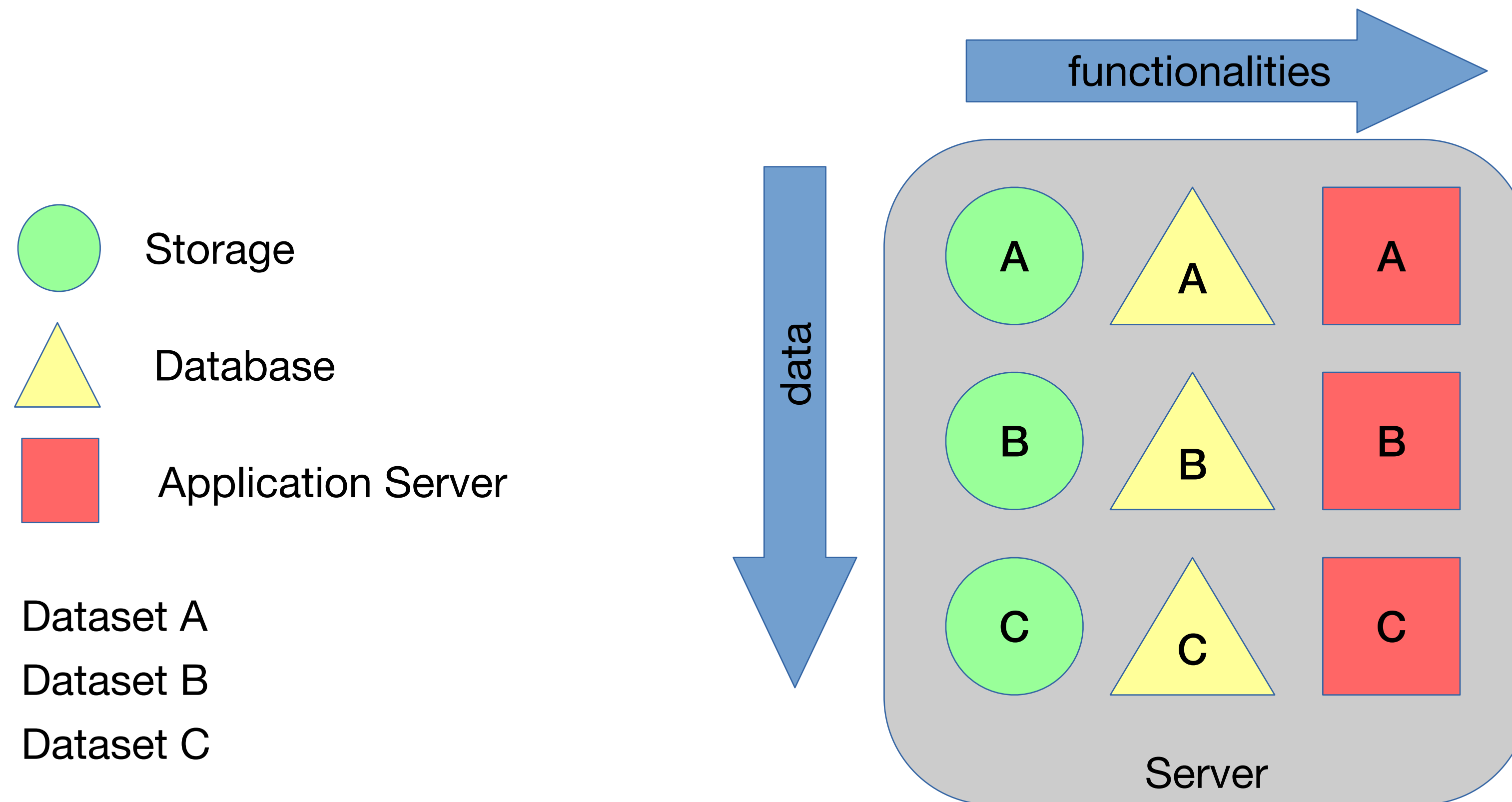
● Main concerns: **why distributed systems?**

- **Modularity**, decoupling different components (concerns) (e.g., storage, database, application server, web server)
- **Performance / Scalability** (more servers doing the work means more speed!)
- **Availability / Dependability** (hardware and software often fail!)

● Main patterns, mechanisms and architectures: **how to distribute?**

Monolithic system

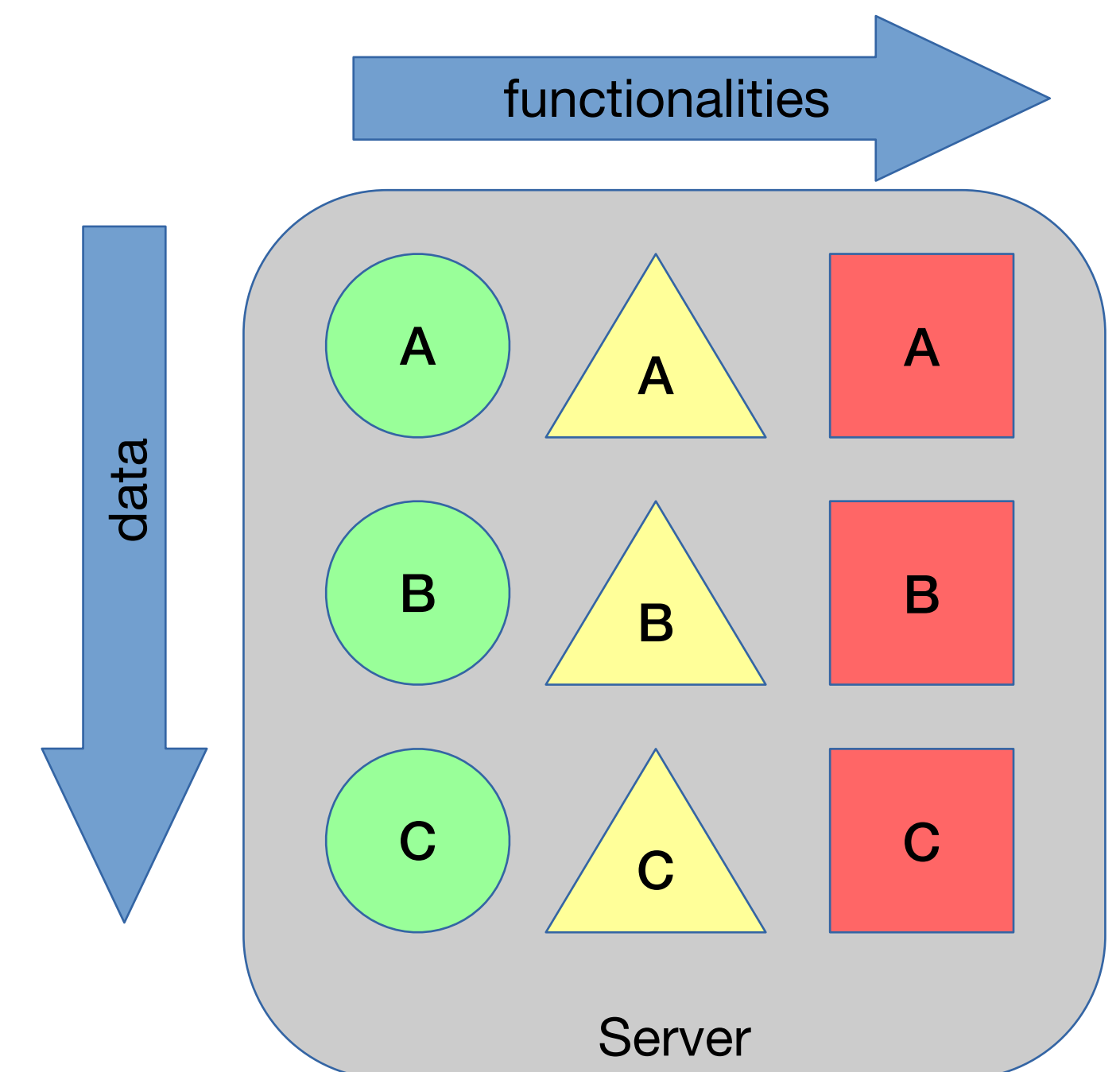
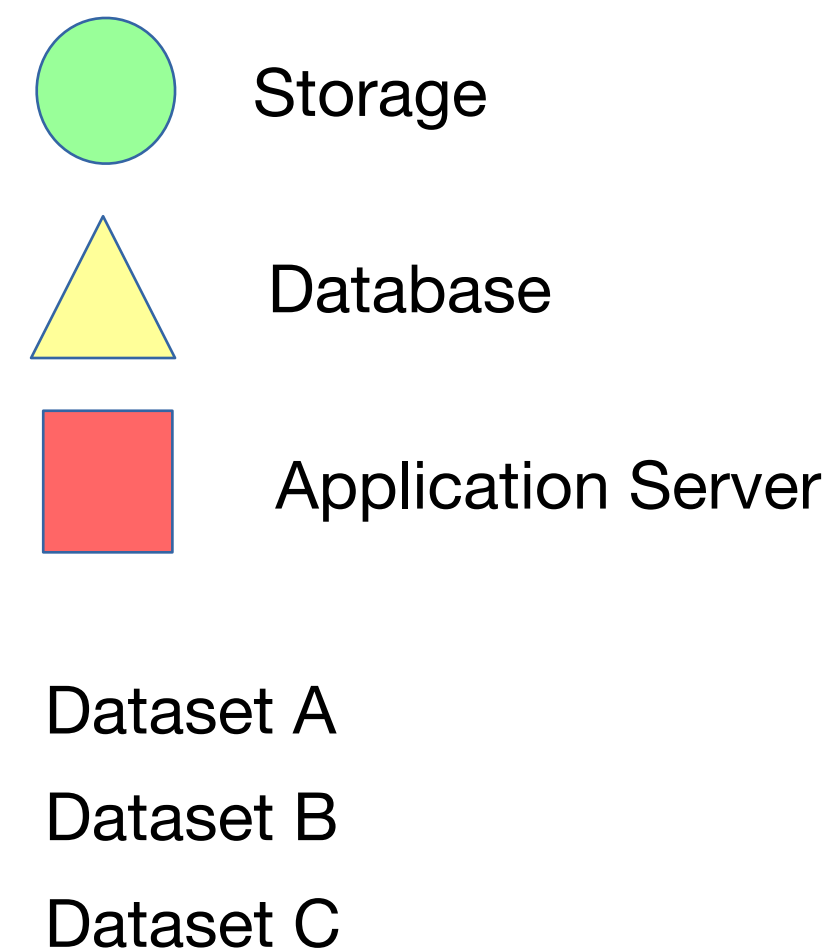
- Multiple components (i.e., storage, database, application server) serving multiple targets (e.g., data from different clients) in the same server



Challenges

◎ **Question:** Any ideas on how to solve these two challenges?

- What if the server cannot handle the load imposed by clients?
- What if the server fails?



Challenges

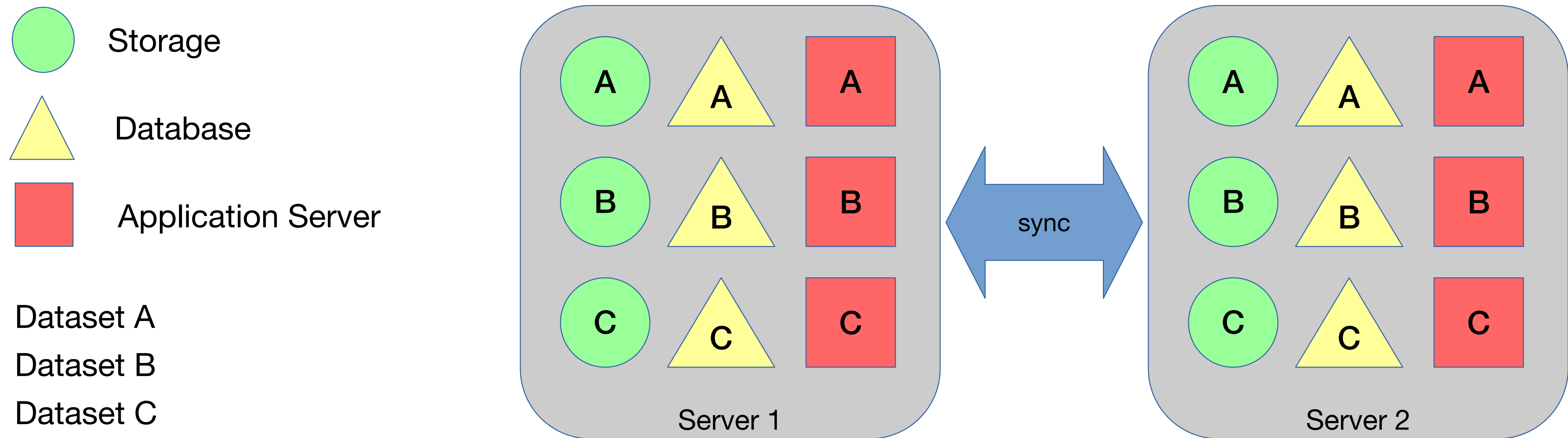
- ◎ **Question:** Any ideas on how to solve these two challenges?
 - What if the server cannot handle the load imposed by clients?
 - What if the server fails?
- ◎ What if the server cannot handle the load imposed by clients?
 - **Scale-up:** increase server resources
 - **Scale-out:** increase number of servers
- ◎ What if the server fails?
 - **Redundancy:** have redundant servers

Distributed system

- ◎ Main distribution patterns/mechanisms:
 - ▶ **Replication**
 - ▶ **Partitioning**
 - ▶ **Service-orientation**
- ◎ All of these address the **scale-out** of a service/application
- ◎ Replication also provides **redundancy!**
- ◎ Not mutually exclusive, can be combined

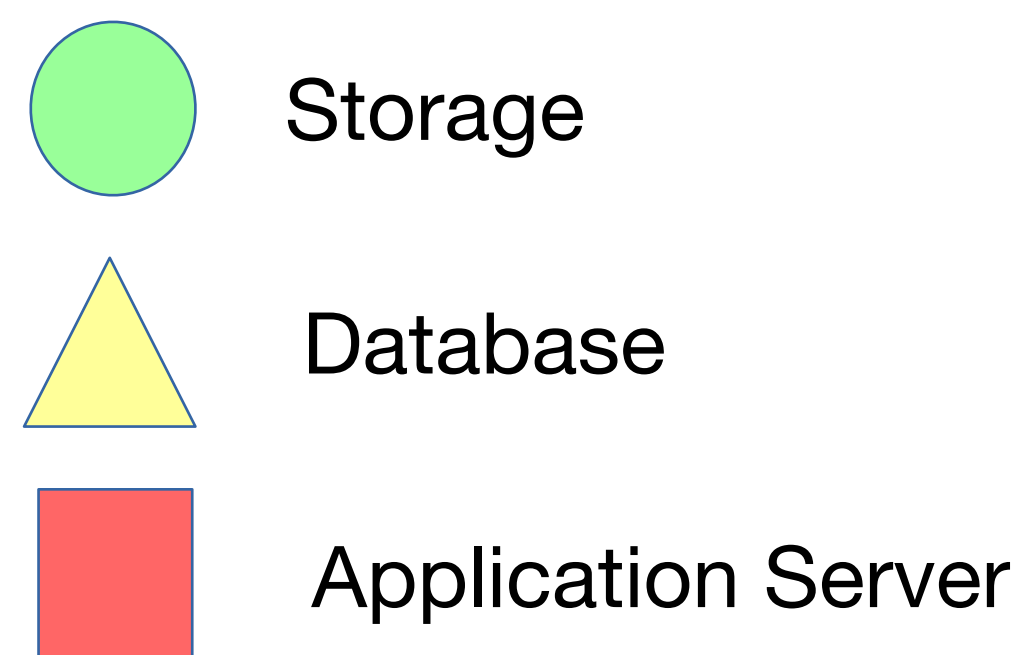
Replication

- Multiple copies of the same data and/or functionality
- Addresses **dependability** (i.e., if Server 1 fails, Server 2 does its work) and **scale-out** (i.e., user “read” requests balanced across Servers 1 and 2)

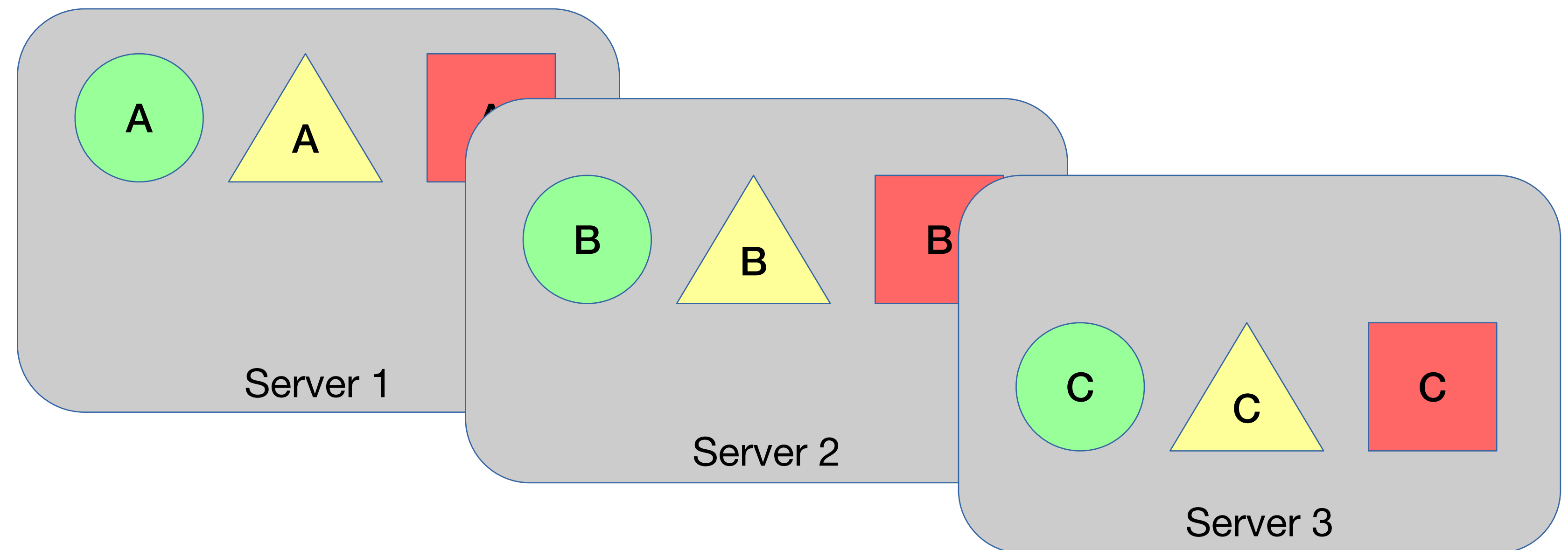


Partitioning

- A server is split horizontally (*Sharding*)
 - It can be applied to computation (functionality) and/or data
- Addresses **scale-out** (i.e., Server 1 handles data/computation for client A, Server 2 for client B, and so on...)

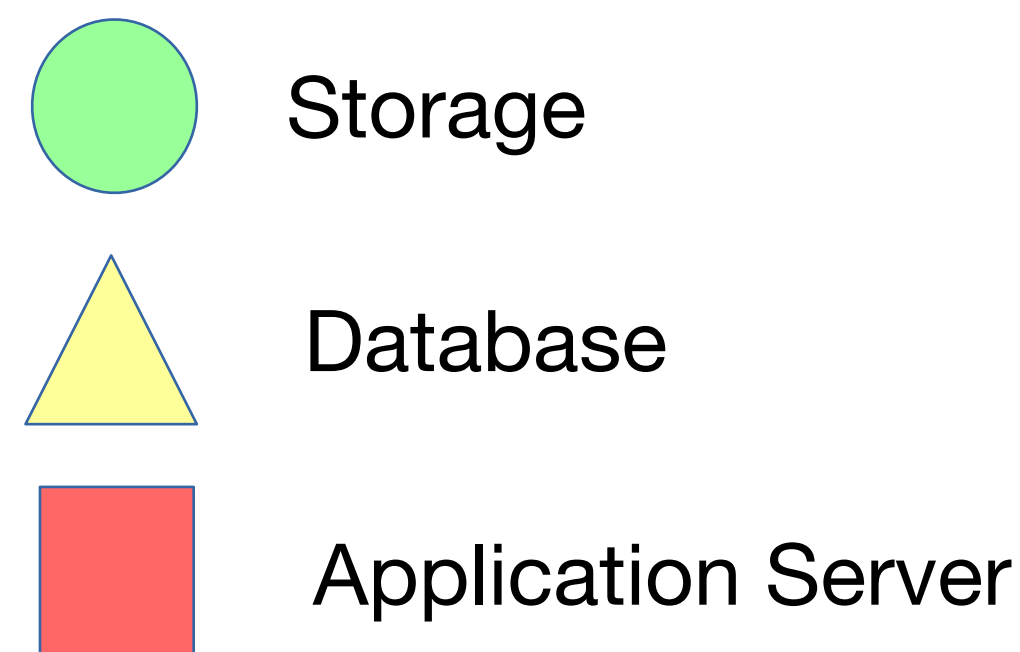


Dataset A
Dataset B
Dataset C

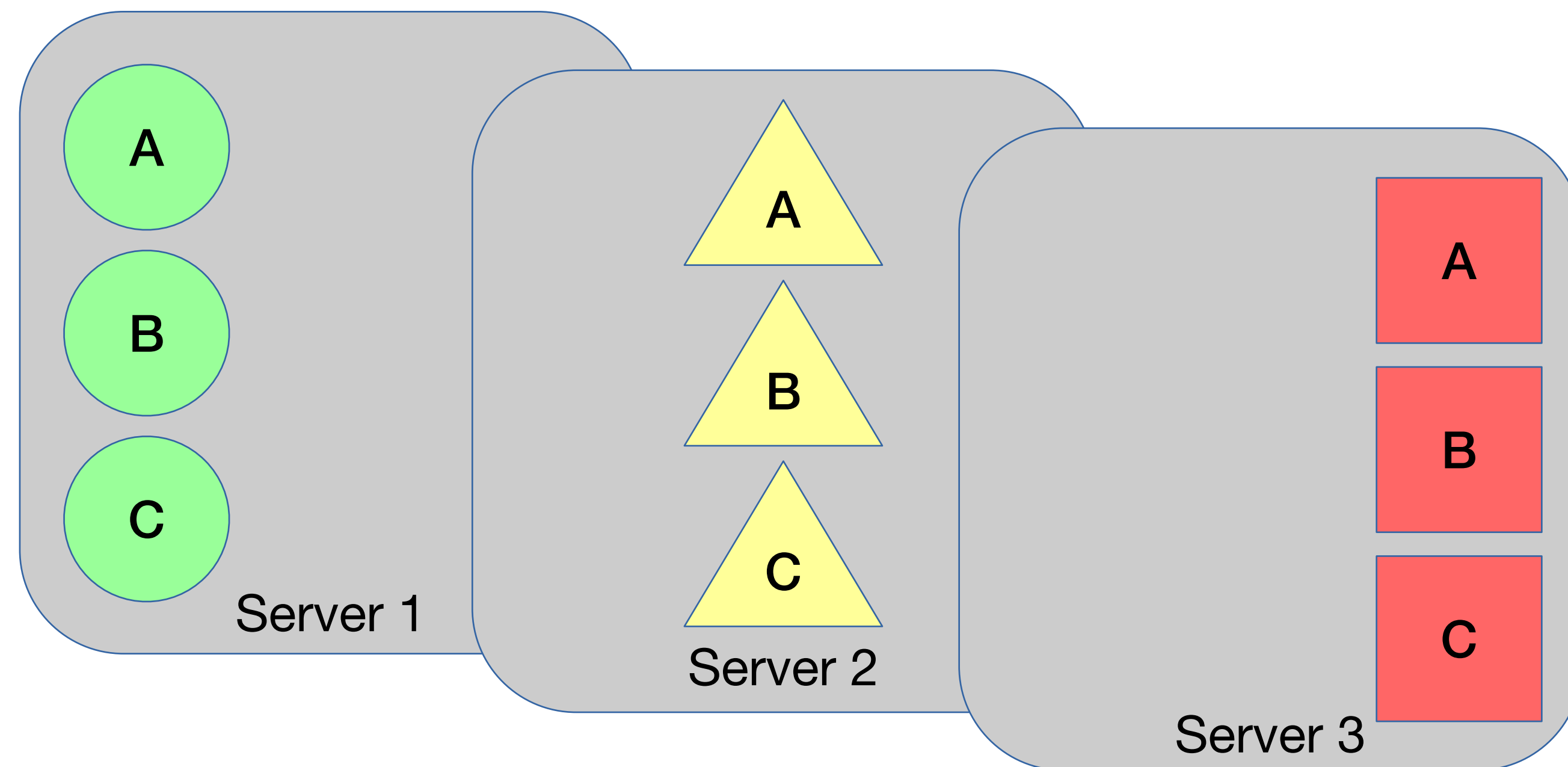


Service-Oriented Architecture (SOA)

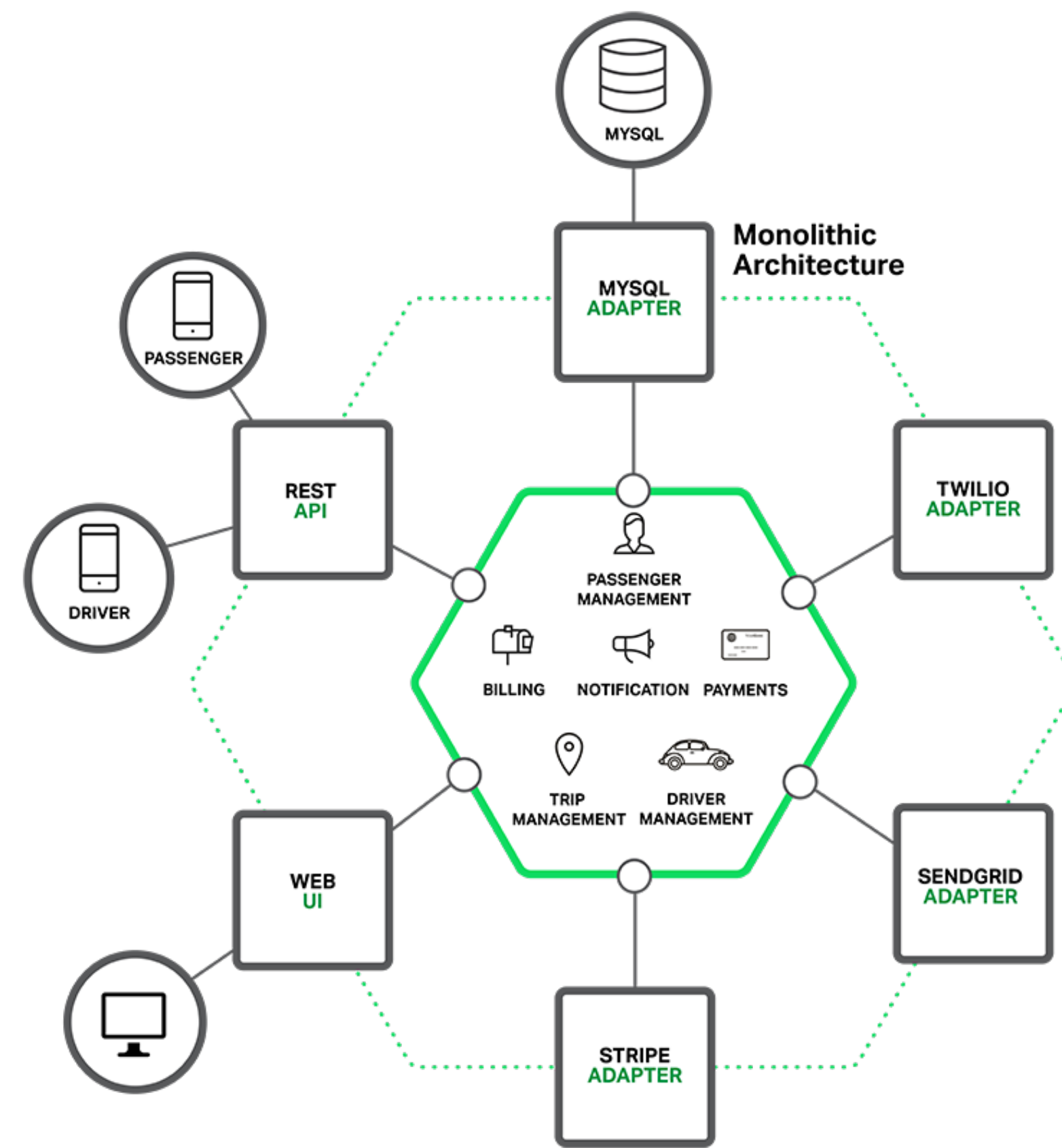
- A server is split vertically (e.g., Microservices)
- Addresses **scale-out** and **modularity** (i.e., Server 1 handles storage services, Server 2 handles database services and Server 3 handles application services)



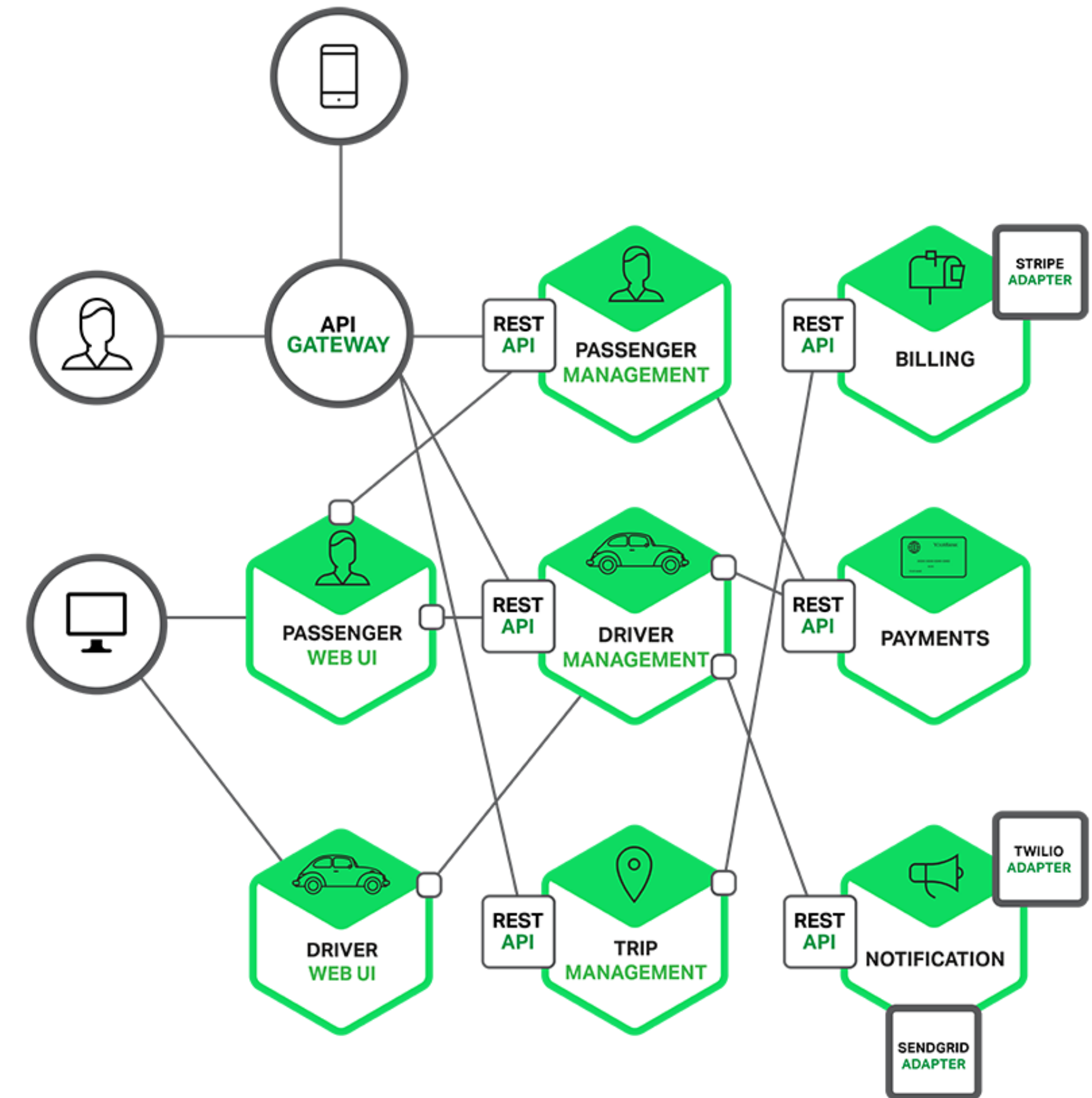
Dataset A
Dataset B
Dataset C



Monolithic to Microservices



Monolithic

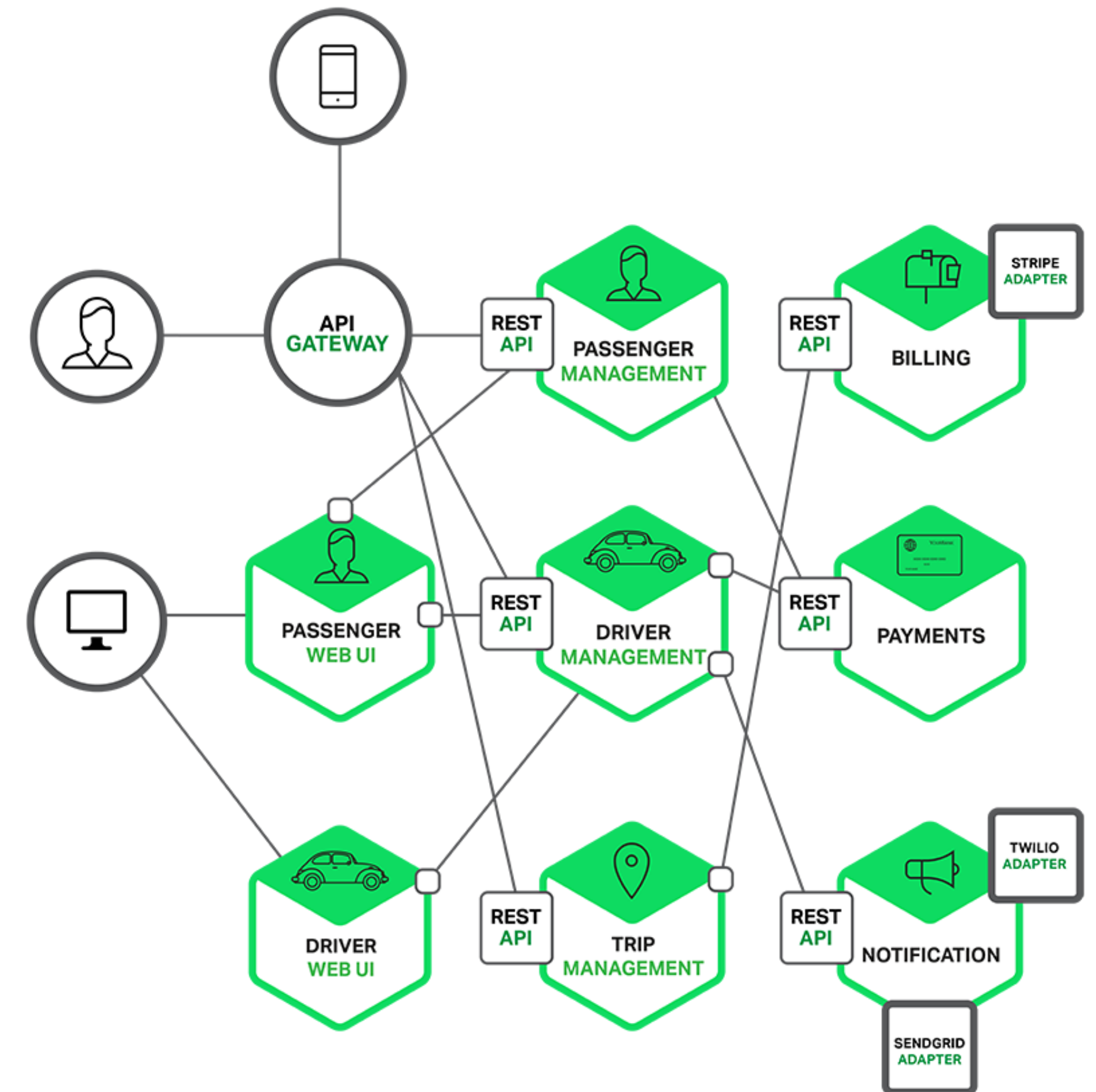


Microservices

Source: <https://www.nginx.com/blog/introduction-to-microservices/>

Microservices

- Each service implements specific **functionality** and can **scale independently**
- Decomposition may be troublesome:
 - how micro is micro? Should I further decompose my functionalities into smaller services?
- Consistency
 - How can one ensure data is consistent across all services?
- Complex deployment and testing



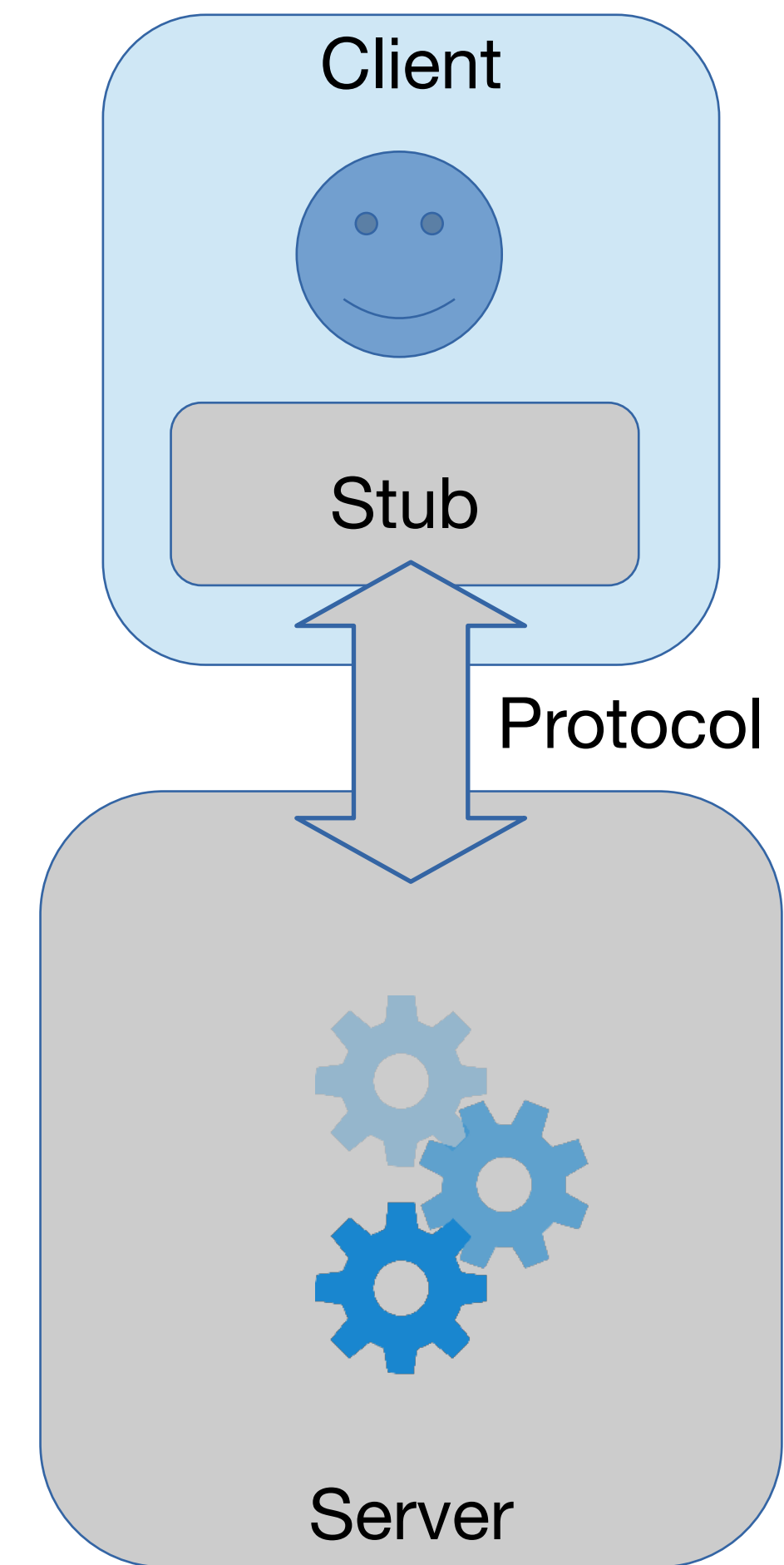
Microservices

Distributed architectures

- © We talked about distribution mechanisms and patterns, what about the **architectures** that enable these?

Client-server

- **Functionality** and **data** are in the server
- A **stub** runs embedded in the client
 - provides an API to interact with the server
 - abstracts the details of the protocol
 - It is part of the server software package
- Example: the Web
 - “protocol” is HTTP
- If there is only one server...
 - One does not grant scalability or dependability!



Proxy-server

● The **proxy** abstracts the interaction with multiple servers

- Good for implementing transparent **replication and/or sharding** of data and functionality!

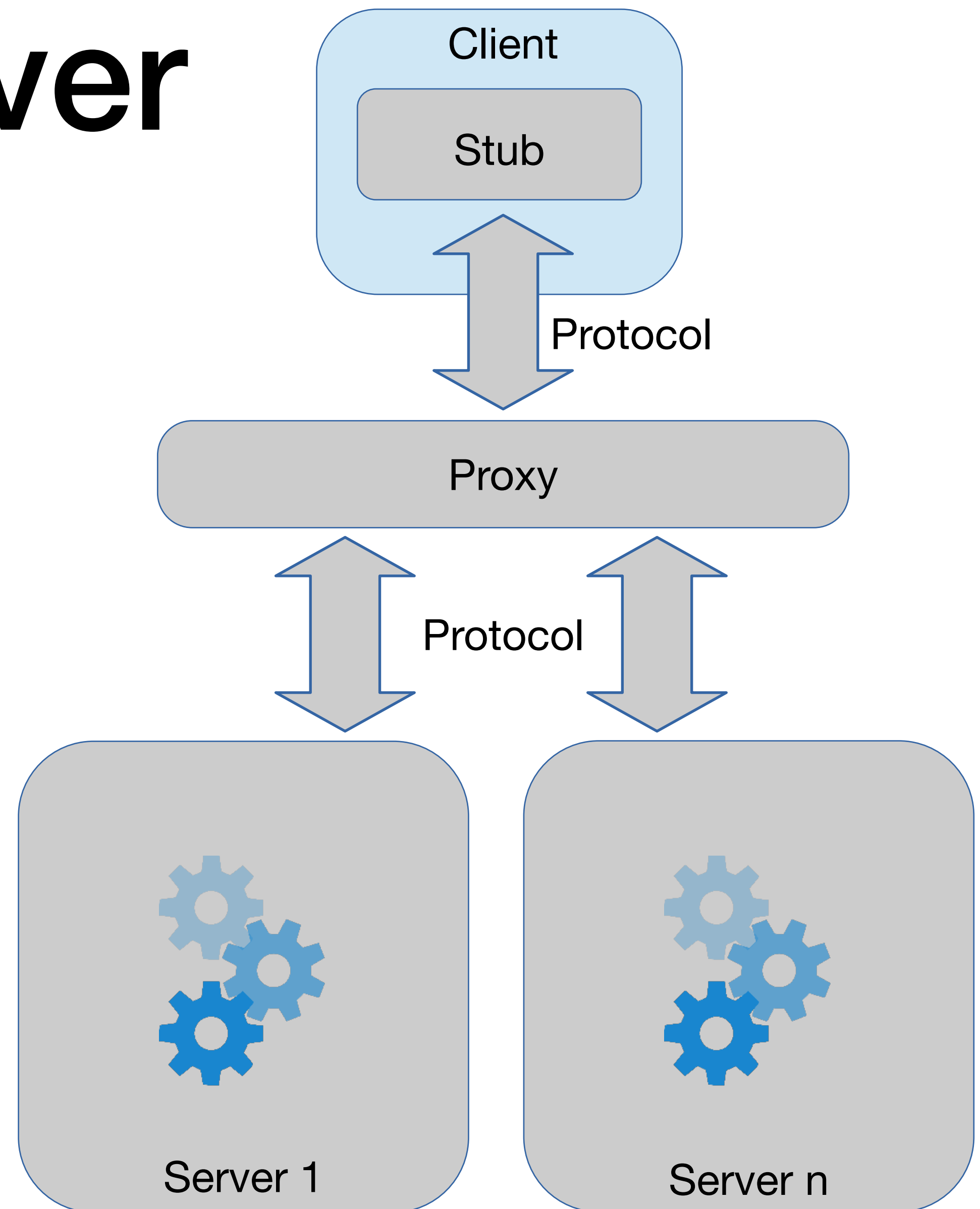
● A **proxy** is different from a **stub**!

- **Proxy**: makes the underlying servers transparent to clients, sitting below the protocol to communicate with clients
- **Stub**: makes the protocol transparent to clients
- They can be combined!

● The proxy may be a **scalability** and **availability bottleneck**!

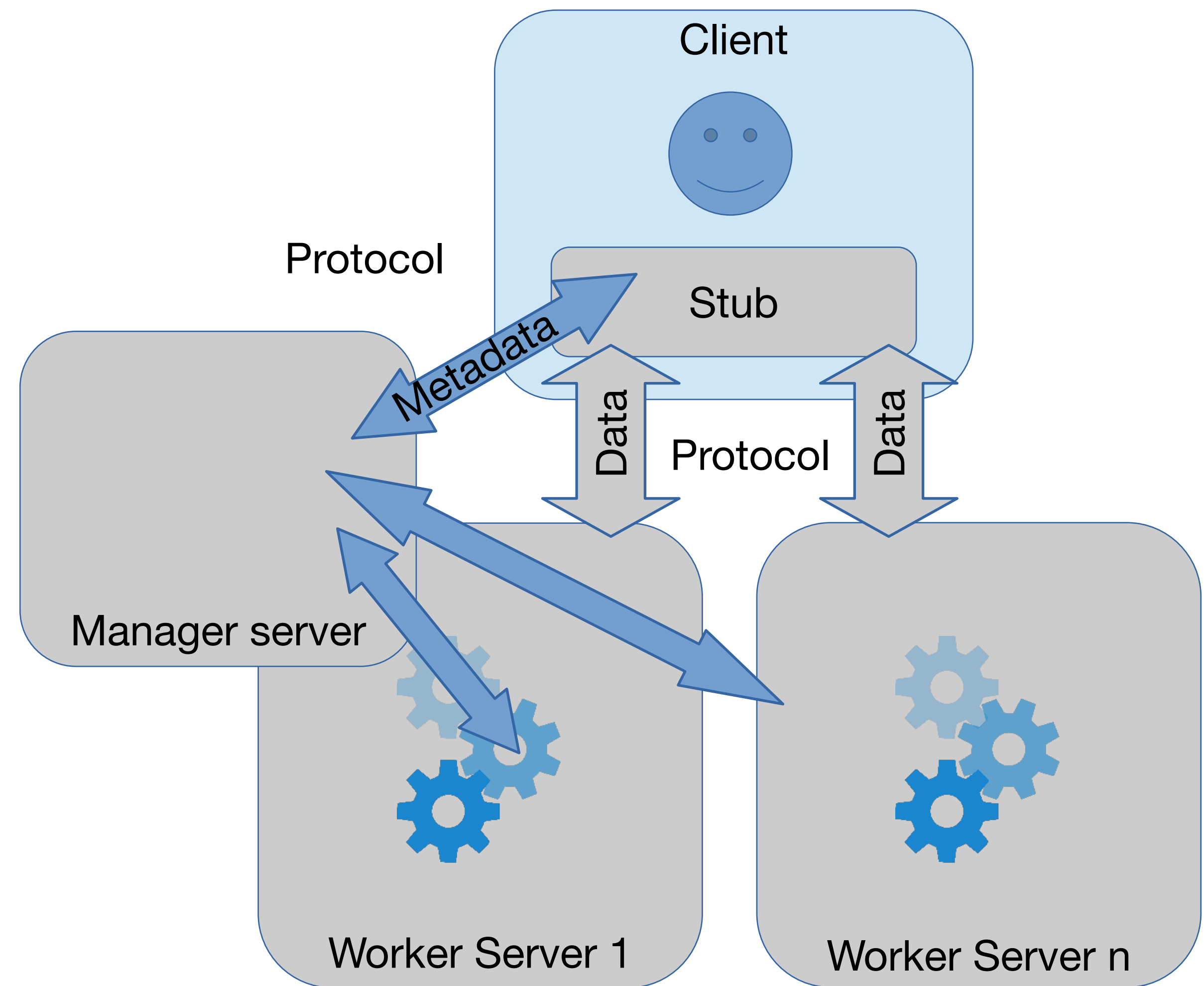
- Single point of contention and failure!

● Example: MongoDB



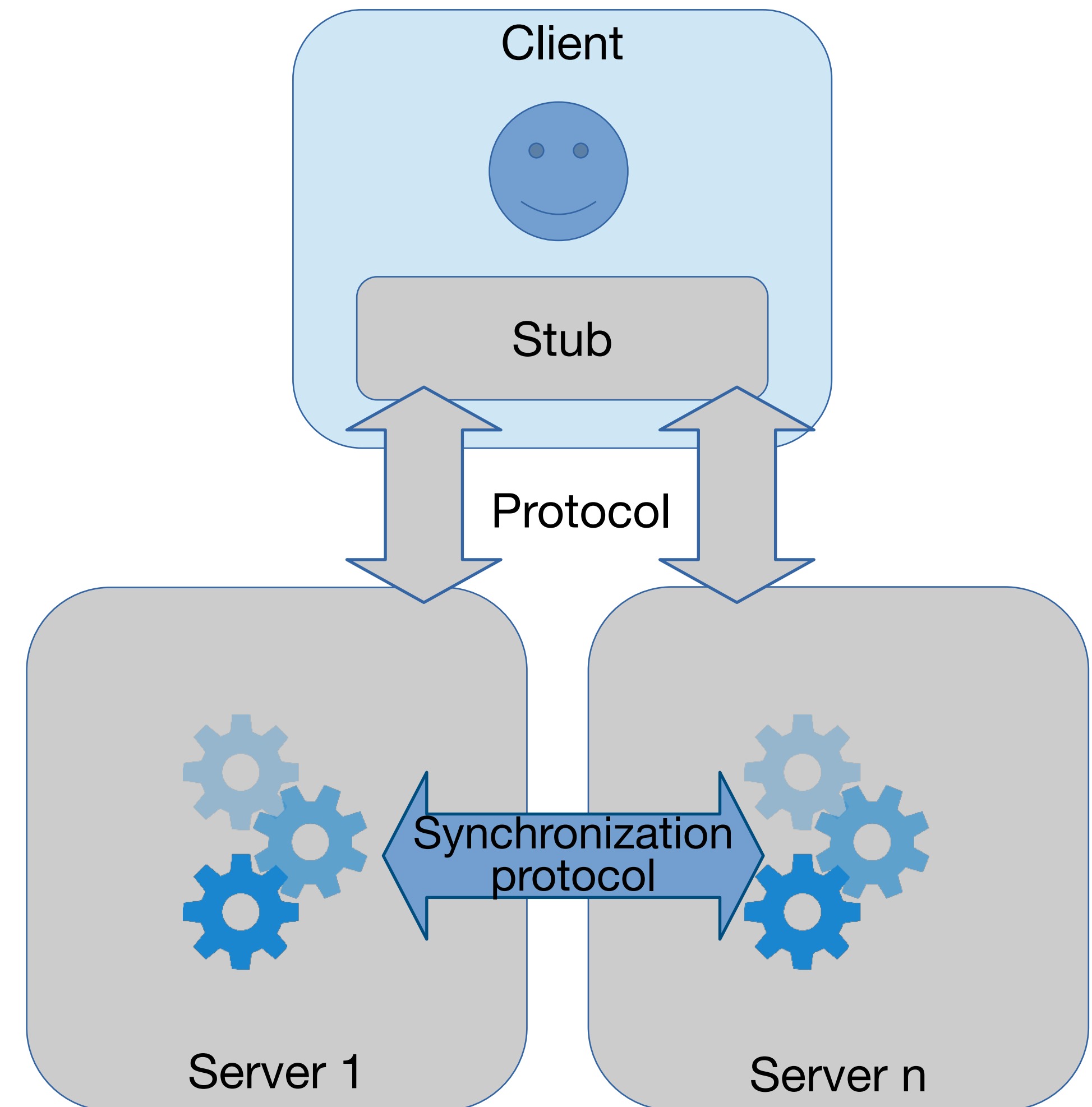
Manager-worker

- Functionality is split between a **manager** and **worker** servers
- Common architecture, for example, for storage systems such as HDFS, Lustre, ...
 - **Manager:** handles the location of files (metadata)
 - **Workers:** handle the content of files (data)
 - Data may be **replicated/sharded** across several workers
- Further addresses **scale-out** i.e., by splitting functionality!
- Manager may still be a single point of contention and failure!



Server group

- All servers can serve/process client requests
 - The failure of one server does not compromise any **data/functionality** of the service!
- A **synchronization protocol** is used to ensure the replication/consistency of servers' state
 - Requires **coordination** across servers
- **No single point of failures!**
- Coordination across server makes it **harder to scale-out!**
- Example: Zookeeper



Bus

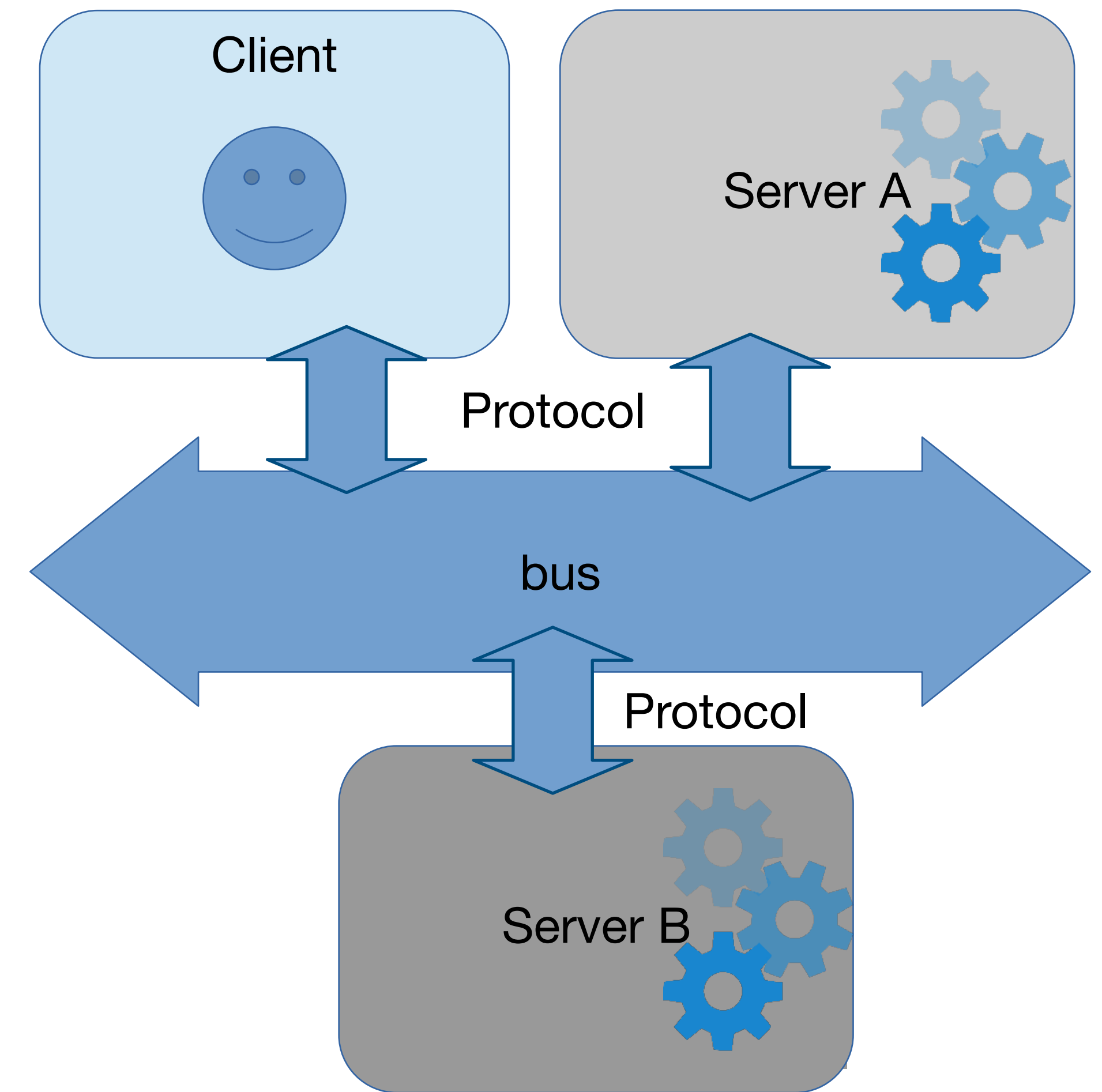
- A **bus** routes messages

- Participants **publish and consume** messages from the bus by using a given protocol
- Decouples producers from consumers

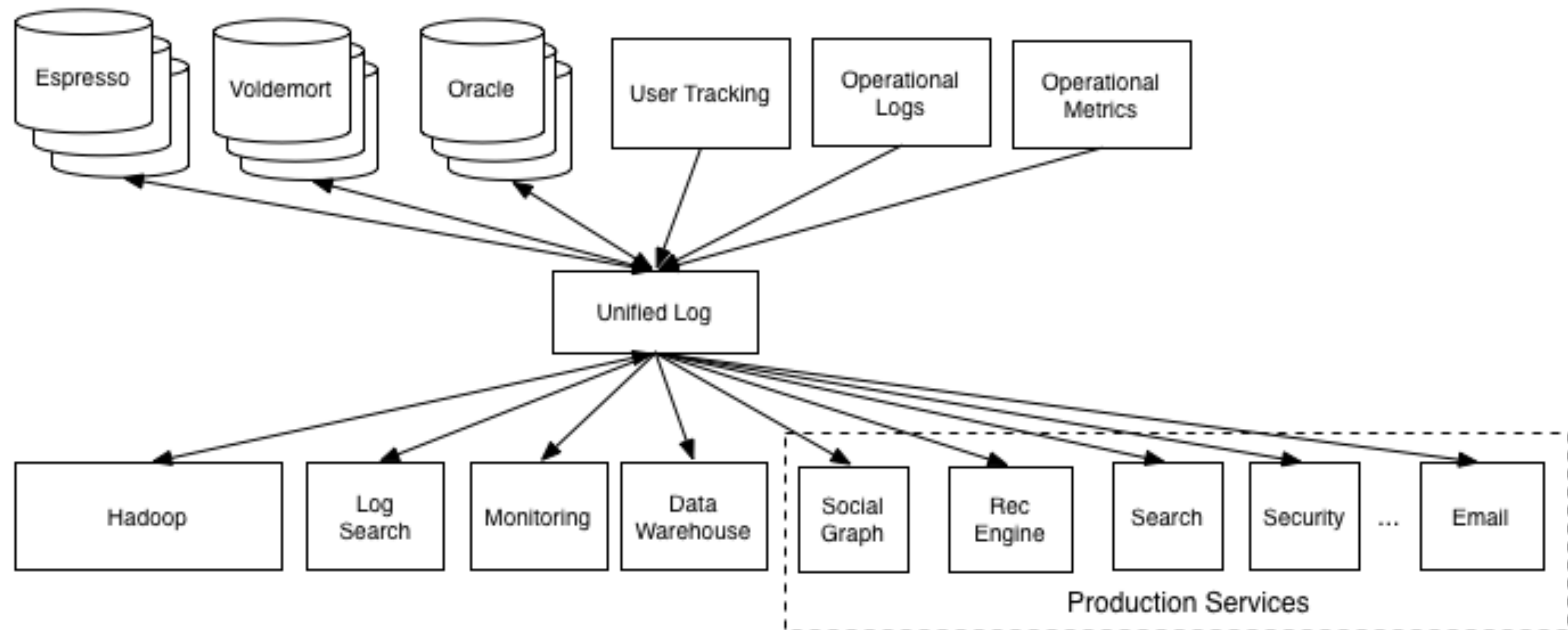
- Increased **flexibility**!

- Clients (users/servers) can **both consume** a set of messages and **produce** another set
- Easier to add new clients to the **bus**

- Example: Kafka

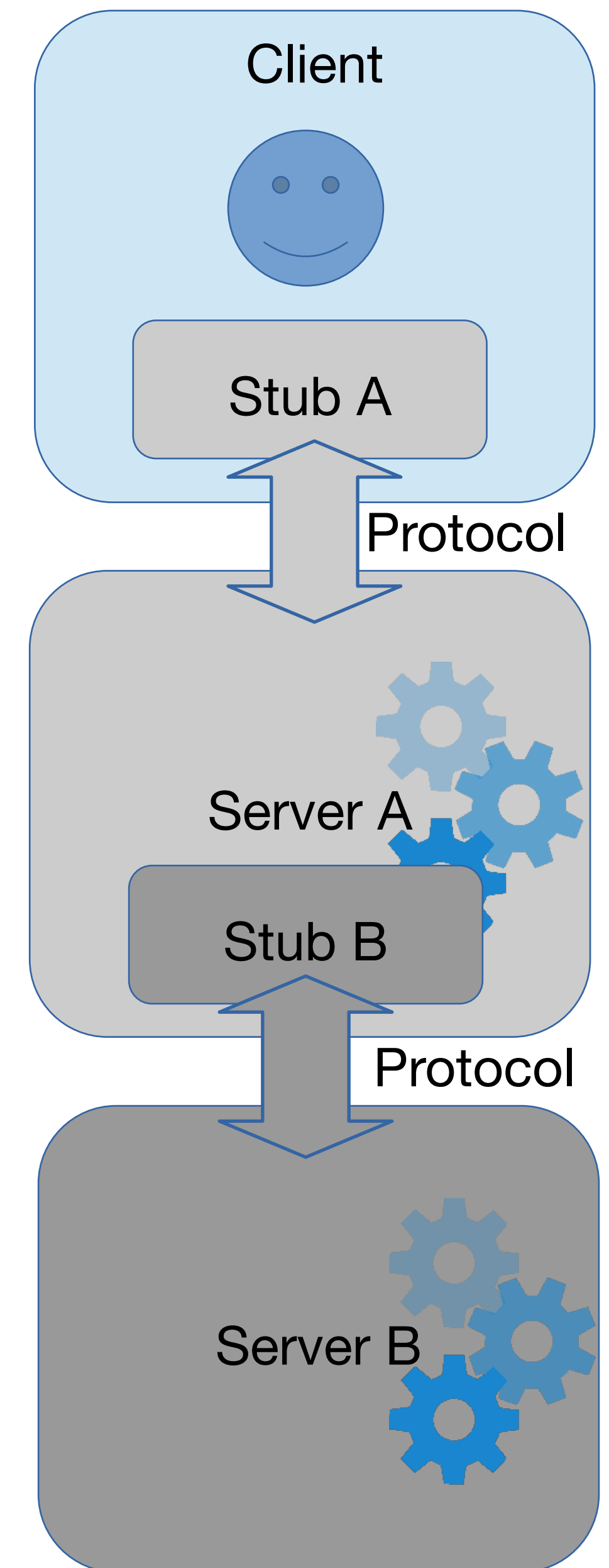


Apache Kafka



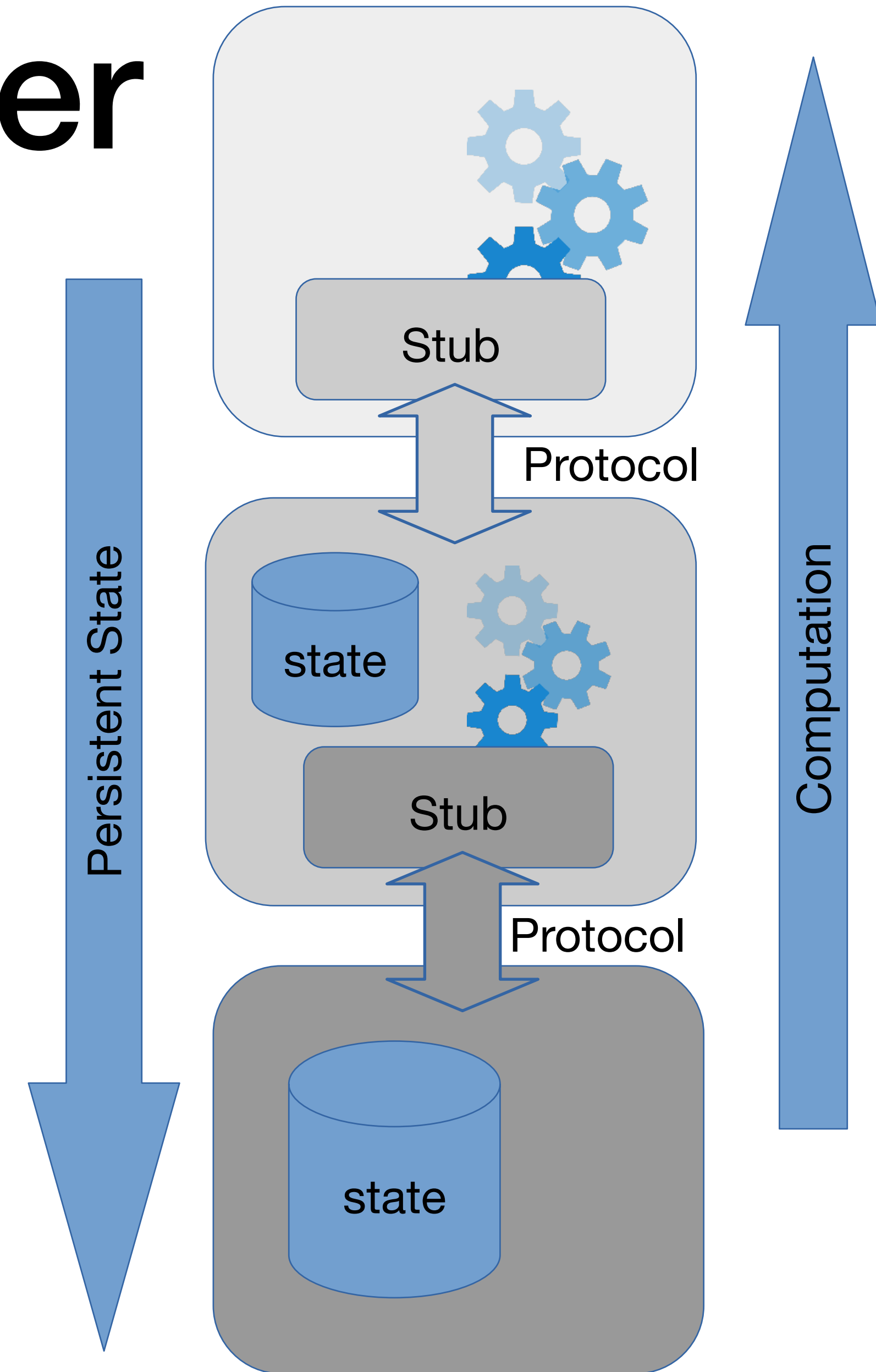
Multi-tier

- Each server acts as a client of the next tier
 - Nested client-server pattern
- Allows independent deployment and scaling of different functionality
- Example: Swap
 - “protocol A” == Web (e.g.)
 - “Stub B” == Database Driver!
 - “protocol B” uses SQL



State in multi-tier

- Typically (not always the case...)
 - No state in upper tiers: Web server
 - Transient / cached state in middle tiers: Application Server
 - Persistent state at lower tiers: Database
- **Question:** Which tiers does one need to consider to ensure availability?
- Computation is typically easier to replicate and shard than persistent state
 - **Question:** Based on what we discussed so far, any idea on why?

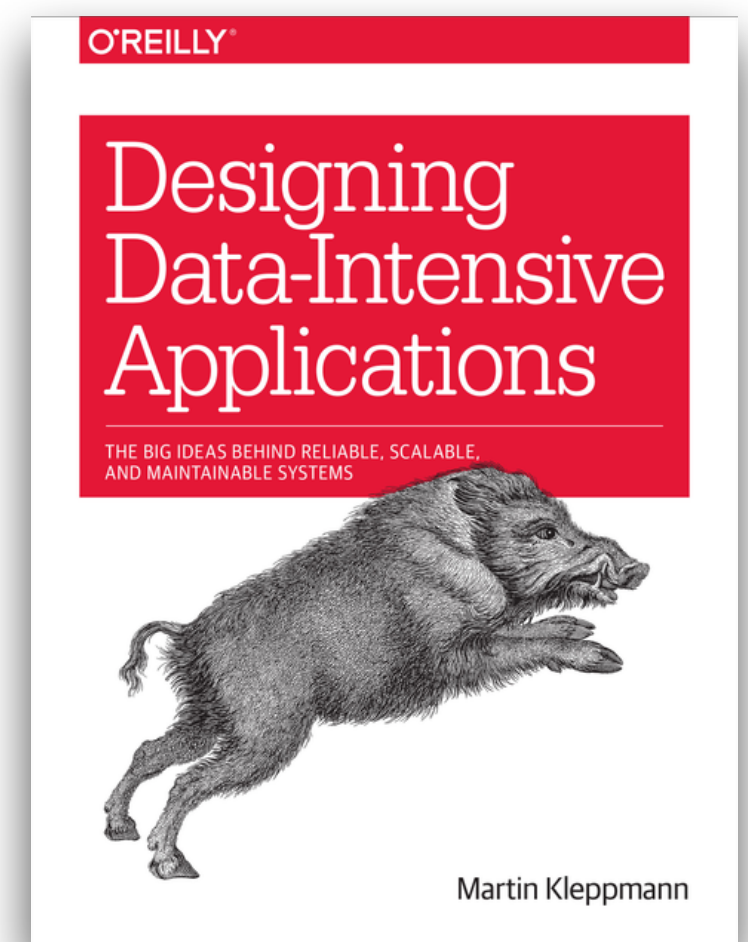


Final Remarks

- Distribution **patterns**, **mechanisms** and **architectures** can be combined to meet different performance, scale, and dependability requirements
- Examples:
 - Data can be sharded across several servers for scalability, while data shards can be replicated for dependability purposes
 - Proxy and manager server components may follow a server-group architecture to avoid being single point of failures and/or contention
 - Each layer of a multi-tier application may follow a specific architecture and implement its own distribution mechanisms

Further Reading

- M. Kleppmann. *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*. O'Reilly, 2017
- C. Tang, T. Kooburat, P. Venkatachalam, A. Chander, Z. Wen, A. Narayanan, P. Dowell, and R. Karl. 2015. *Holistic configuration management at Facebook*. In Proceedings of the 25th Symposium on Operating Systems Principles (SOSP '15). ACM, New York, NY, USA, 328-343.



Questions?